Resit Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

July 16, 2026

- This exam consists of Exercises **1–9 for 202001172** and **1–11 for 202001171**.
- You have **three hours** to solve the exam.
- You must **motivate** and **justify your answers clearly**. Answers without explanation will not receive credit.
- The use of calculators or any other electronic devices is not permitted.
- A formula sheet is attached to this exam. No other course materials are allowed.

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1. Evaluate the limit:
$$\lim_{x\to\infty}\left(\sqrt{x^2+6x}-x\right).$$

2. The functions f and g, together with their derivatives, take the following values:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	4	2	3	-1

Find the derivatives with respect to x of the following expressions, evaluated at $x=2$:

(a) $\dfrac{\sqrt{f(x)}}{g(x)}$; (b) $\ln(f(g(x)))$.

3. Evaluate the following integrals:

(a) $\displaystyle\int_e^{\infty}\frac{1}{x(\ln x)^2}\,dx$; (b) $\displaystyle\int_0^{\pi^2/4}\cos(\sqrt{x})\,dx$.

4. Let the function f be given by
$$f(x,y)=\sqrt{x+e^{2y}}.$$
Find an equation of the tangent plane to the graph of f at the point $(3,0,2)$.

5. Let C be the circle with center $(-2,3)$ that is described by the polar equation
$$r^2+ar\cos\theta-6r\sin\theta=-4,$$
where $a\in\mathbb{R}$ is a constant.
   (a) Find a.
   (b) Find the radius of circle C.

6. Show that the limit
$$\lim_{(x,y)\to(0,0)}\frac{3y(x+y)}{x^2+y^2}$$
does **not** exist.

7. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function with

$$\frac{\partial f}{\partial x}(x, y) = 4x \quad \text{and} \quad \frac{\partial f}{\partial y}(x, y) = 2y.$$

If $x(t) = \ln t$ and $y(t) = t^3$, compute $\dfrac{d}{dt} f(x(t), y(t))$ for $t > 0$.

8. Consider the function

$$f(x, y) = \frac{1}{3}x^3 + y^2 + 2xy - 3x.$$

(a) Find all critical points of f.

(b) Determine the absolute minimum of f on the line segment defined by $y = 0$ and $-1 \leq x \leq 3$.

9. Find the solution $y = y(x)$ to the following initial value problem:

$$\begin{cases} x^2 y' + 2xy = e^x, \quad x > 0, \\ y(1) = e. \end{cases}$$

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10. Find the six complex solutions z (in polar form) of the following equation:

$$z^6 - 3z^3 - 4 = 0.$$

11. Find the solution $y = y(x)$ to the following initial value problem:

$$y'' + 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

Point distribution:

Exercise	1	2		3		4	5		6	7	8		9	10	11	Total
		a	b	a	b		a	b			a	b				
202001171	4	2	2	2	2	4	2	2	4	4	2	2	4	5	4	45
202001172	4	2	2	2	2	4	2	2	4	4	2	2	4	–	–	36

Exam grade (202001171): $1 + (\text{obtained points}) \times \frac{1}{5}$

Exam grade (202001172): $1 + (\text{obtained points}) \times \frac{1}{4}$

Solutions to Resit Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

July 6, 2026

1. **(4 points)** Evaluate the limit:

$$\lim_{x\to\infty}\left(\sqrt{x^2+6x}-x\right).$$

Solution:

This is an indeterminate form of type $\infty-\infty$. We multiply by the conjugate:

$$\begin{aligned}\lim_{x\to\infty}\left(\sqrt{x^2+6x}-x\right) &= \lim_{x\to\infty}\left(\sqrt{x^2+6x}-x\right)\cdot\frac{\sqrt{x^2+6x}+x}{\sqrt{x^2+6x}+x}\\ &= \lim_{x\to\infty}\frac{(x^2+6x)-x^2}{\sqrt{x^2+6x}+x}\\ &= \lim_{x\to\infty}\frac{6x}{\sqrt{x^2+6x}+x}\end{aligned}$$

For $x>0$, divide numerator and denominator by x:

$$= \lim_{x\to\infty}\frac{6}{\sqrt{1+6/x}+1} = \frac{6}{\sqrt{1+0}+1} = \frac{6}{2} = 3$$

Answer: $\boxed{3}$

2. **(2+2 = 4 points)** The functions f and g, together with their derivatives, take the following values:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	4	2	3	-1

Find the derivatives with respect to x of the following expressions, evaluated at $x=2$:

(a) $\dfrac{\sqrt{f(x)}}{g(x)}$; (b) $\ln(f(g(x)))$.

Solution:

(a) Using the quotient rule and chain rule:

$$\begin{aligned}\frac{d}{dx}\left[\frac{\sqrt{f(x)}}{g(x)}\right] &= \frac{g(x)\cdot\frac{d}{dx}[\sqrt{f(x)}]-\sqrt{f(x)}\cdot g'(x)}{[g(x)]^2}\\ &= \frac{g(x)\cdot\frac{f'(x)}{2\sqrt{f(x)}}-\sqrt{f(x)}\cdot g'(x)}{[g(x)]^2}\end{aligned}$$

At $x=2$:

$$= \frac{2\cdot\frac{3}{2\sqrt{4}}-\sqrt{4}\cdot(-1)}{2^2} = \frac{\frac{3}{2}+2}{4} = \frac{7}{8}$$

Answer: $\boxed{\dfrac{7}{8}}$

(b) Using the chain rule twice:

$$\frac{d}{dx}[\ln(f(g(x)))] = \frac{1}{f(g(x))} \cdot f'(g(x)) \cdot g'(x)$$

At $x = 2$, we have $g(2) = 2$, so:

$$= \frac{1}{f(g(2))} \cdot f'(g(2)) \cdot g'(2) = \frac{1}{f(2)} \cdot f'(2) \cdot g'(2)$$
$$= \frac{1}{4} \cdot 3 \cdot (-1) = -\frac{3}{4}$$

Answer: $\boxed{-\frac{3}{4}}$

3. **(2+2 = 4 points)** Evaluate the following integrals:

(a) $\displaystyle\int_e^\infty \frac{1}{x(\ln x)^2}\,dx$; (b) $\displaystyle\int_0^{\pi^2/4} \cos(\sqrt{x})\,dx$.

Solution:

(a) Use substitution: Let $u = \ln x$, then $du = \frac{1}{x}dx$
When $x = e$: $u = 1$; when $x \to \infty$: $u \to \infty$

$$\lim_{a\to\infty}\int_e^a \frac{1}{x(\ln x)^2}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{u^2}\,du = \lim_{b\to\infty}\left[-\frac{1}{u}\right]_1^b = \lim_{b\to\infty}\left(-\frac{1}{b}+1\right) = 1$$

Answer: $\boxed{1}$ (the integral converges)

(b) Use substitution: Let $u = \sqrt{x}$, then $x = u^2$ and $dx = 2u\,du$
When $x = 0$: $u = 0$; when $x = \pi^2/4$: $u = \pi/2$

$$\int_0^{\pi^2/4} \cos(\sqrt{x})\,dx = \int_0^{\pi/2} \cos(u) \cdot 2u\,du = 2\int_0^{\pi/2} u\cos(u)\,du$$

Using integration by parts with $v = u$, $dw = \cos(u)du$, so $dv = du$, $w = \sin(u)$:

$$= 2\left[u\sin(u)\Big|_0^{\pi/2} - \int_0^{\pi/2}\sin(u)\,du\right]$$
$$= 2\left[\frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) - 0 + \cos(u)\Big|_0^{\pi/2}\right]$$
$$= 2\left[\frac{\pi}{2} + \cos\left(\frac{\pi}{2}\right) - \cos(0)\right]$$
$$= 2\left[\frac{\pi}{2} + 0 - 1\right] = \pi - 2$$

Answer: $\boxed{\pi - 2}$

4. **(4 points)** Let the function f be given by

$$f(x, y) = \sqrt{x + e^{2y}}.$$

Find an equation of the tangent plane to the graph of f at the point $(3, 0, 2)$.

Solution:

Find partial derivatives:

$$\frac{\partial f}{\partial x} = \frac{1}{2\sqrt{x+e^{2y}}} \cdot 1 = \frac{1}{2\sqrt{x+e^{2y}}}$$
$$\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{x+e^{2y}}} \cdot 2e^{2y} = \frac{e^{2y}}{\sqrt{x+e^{2y}}}$$

At $(3,0)$:

$$f_x(3,0) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$
$$f_y(3,0) = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

The tangent plane equation is:

$$z - z_0 = f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$
$$z - 2 = \frac{1}{4}(x-3) + \frac{1}{2}(y-0)$$
$$z = \frac{1}{4}x - \frac{3}{4} + \frac{1}{2}y + 2$$
$$z = \frac{1}{4}x + \frac{1}{2}y + \frac{5}{4}$$

Answer: $\boxed{z = \frac{1}{4}x + \frac{1}{2}y + \frac{5}{4}}$ or equivalently $\boxed{x + 2y - 4z + 5 = 0}$

5. **(2+2 = 4 points)** Let C be the circle with center $(-2,3)$ that is described by the polar equation

$$r^2 + ar\cos\theta - 6r\sin\theta = -4,$$

where $a \in \mathbb{R}$ is a constant.

(a) Find a.

(b) Find the radius of circle C.

Solution:

(a) Convert the polar equation to Cartesian using $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$:

$$x^2 + y^2 + ax - 6y = -4$$
$$(x^2 + ax) + (y^2 - 6y) = -4$$

Complete the square:

$$\left(x^2 + ax + \frac{a^2}{4}\right) - \frac{a^2}{4} + \left(y^2 - 6y + 9\right) - 9 = -4$$
$$\left(x + \frac{a}{2}\right)^2 + (y-3)^2 = -4 + \frac{a^2}{4} + 9 = 5 + \frac{a^2}{4}$$

The center is at $\left(-\frac{a}{2}, 3\right)$. Given that the center is $(-2,3)$:

$$-\frac{a}{2} = -2 \implies a = 4$$

Answer: $\boxed{a = 4}$

(b) With $a = 4$:

$$(x+2)^2 + (y-3)^2 = 5 + \frac{16}{4} = 5 + 4 = 9$$

The radius is $r = \sqrt{9} = 3$.

Answer: $\boxed{3}$

6. **(4 points)** Show that the limit

$$\lim_{(x,y)\to(0,0)} \frac{3y(x+y)}{x^2+y^2}$$

does **not** exist.

Solution:

We'll approach $(0,0)$ along different paths and show we get different limits.

Path 1: Along $y = 0$

$$\lim_{x\to 0} \frac{3 \cdot 0 \cdot (x+0)}{x^2+0} = \lim_{x\to 0} \frac{0}{x^2} = 0$$

Path 2: Along $x = 0$

$$\lim_{y\to 0} \frac{3y(0+y)}{0+y^2} = \lim_{y\to 0} \frac{3y^2}{y^2} = 3$$

Since we obtain limit 0 along $y = 0$ but limit 3 along $x = 0$, the limit does not exist.

Conclusion: The limit does not exist because different paths yield different values (0 versus 3).

7. **(4 points)** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function with

$$\frac{\partial f}{\partial x}(x,y) = 4x \quad \text{and} \quad \frac{\partial f}{\partial y}(x,y) = 2y.$$

If $x(t) = \ln t$ and $y(t) = t^3$, compute $\dfrac{d}{dt} f(x(t), y(t))$ for $t > 0$.

Solution:

Using the multivariable chain rule:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Calculate derivatives:

$$\frac{dx}{dt} = \frac{1}{t}$$
$$\frac{dy}{dt} = 3t^2$$

Substitute:

$$\begin{aligned}\frac{d}{dt} f(x(t), y(t)) &= 4x(t) \cdot \frac{1}{t} + 2y(t) \cdot 3t^2 \\ &= 4(\ln t) \cdot \frac{1}{t} + 2(t^3) \cdot 3t^2 \\ &= \frac{4 \ln t}{t} + 6t^5\end{aligned}$$

Answer: $\boxed{\dfrac{d}{dt} f(x(t), y(t)) = \dfrac{4 \ln t}{t} + 6t^5}$

8. **(2+2 = 4 points)** Consider the function

$$f(x,y) = \frac{1}{3}x^3 + y^2 + 2xy - 3x.$$

(a) Find all critical points of f.

(b) Determine the absolute minimum of f on the line segment defined by $y = 0$ and $-1 \leq x \leq 3$.

Solution:

(a) Find partial derivatives:

$$f_x = x^2 + 2y - 3$$
$$f_y = 2y + 2x$$

Set both equal to zero:

$$\begin{cases} x^2 + 2y - 3 = 0 \\ 2y + 2x = 0 \end{cases}$$

From the second equation: $y = -x$
Substitute into the first:

$$x^2 - 2x - 3 = 0$$
$$(x-3)(x+1) = 0$$

So $x = 3$ or $x = -1$.
For $x = 3$: $y = -3$
For $x = -1$: $y = 1$
Answer: Critical points are $\boxed{(3,-3)}$ and $\boxed{(-1,1)}$

(b) On the constraint $y = 0$:

$$f(x,0) = \frac{1}{3}x^3 - 3x$$

This is a function of x alone on $[-1,3]$.

$$\frac{d}{dx}f(x,0) = x^2 - 3 = 0 \implies x = \pm\sqrt{3}$$

Only $x = \sqrt{3} \in [-1,3]$. Evaluate at the critical point and the endpoints:

$$f(-1,0) = -\frac{1}{3} + 3 = \frac{8}{3}$$
$$f(\sqrt{3},0) = \frac{3\sqrt{3}}{3} - 3\sqrt{3} = \sqrt{3} - 3\sqrt{3} = -2\sqrt{3}$$
$$f(3,0) = 9 - 9 = 0$$

Since $-2\sqrt{3} \approx -3.46$ is the smallest value, the minimum is attained at $(\sqrt{3},0)$.
Answer: The absolute minimum is $\boxed{f(\sqrt{3},0) = -2\sqrt{3}}$

9. **(4 points)** Find the solution $y = y(x)$ to the following initial value problem:

$$\begin{cases} x^2y' + 2xy = e^x, \quad x > 0, \\ y(1) = e. \end{cases}$$

Solution:

Divide by x^2:

$$y' + \frac{2}{x}y = \frac{e^x}{x^2}$$

This is a first-order linear ODE. The integrating factor is:

$$\mu(x) = e^{\int \frac{2}{x}dx} = e^{2\ln|x|} = e^{\ln x^2} = x^2$$

Multiply the equation by x^2:

$$x^2y' + 2xy = e^x$$

The left side is $\frac{d}{dx}[x^2y]$:

$$\frac{d}{dx}[x^2y] = e^x$$

Integrate:

$$x^2y = \int e^x\,dx = e^x + C$$

Therefore:

$$y = \frac{e^x + C}{x^2}$$

Apply initial condition $y(1) = e$:

$$e = \frac{e^1 + C}{1^2} = e + C$$

Thus $C = 0$.

Answer: $\boxed{y(x) = \frac{e^x}{x^2}}$

10. **(5 points)** Find the six complex solutions z (in polar form) of the following equation:

$$z^6 - 3z^3 - 4 = 0.$$

Solution:

Let $w = z^3$. Then:

$$w^2 - 3w - 4 = 0$$

Factoring:

$$(w - 4)(w + 1) = 0$$

So $w = 4$ or $w = -1$.

Case 1: $z^3 = 4$

In polar form: $4 = 4e^{i\cdot 0}$

The three cube roots are:

$$z_k = \sqrt[3]{4} \cdot e^{i(0+2\pi k)/3} = 4^{1/3} e^{i\cdot 2\pi k/3}, \quad k = 0, 1, 2$$

- $z_0 = 4^{1/3} e^{i\cdot 0} = 4^{1/3}$
- $z_1 = 4^{1/3} e^{i\cdot 2\pi/3}$
- $z_2 = 4^{1/3} e^{i\cdot 4\pi/3}$

Case 2: $z^3 = -1$

In polar form: $-1 = e^{i\pi}$

The three cube roots are:

$$z_k = e^{i(\pi+2\pi k)/3}, \quad k = 0, 1, 2$$

- $z_3 = e^{i\pi/3}$
- $z_4 = e^{i\pi}$
- $z_5 = e^{i\cdot 5\pi/3}$

Answer: The six solutions in polar form are:

$$\boxed{z_0 = 4^{1/3}, \quad z_1 = 4^{1/3} e^{i\cdot 2\pi/3}, \quad z_2 = 4^{1/3} e^{i\cdot 4\pi/3}}$$

$$\boxed{z_3 = e^{i\pi/3}, \quad z_4 = e^{i\pi}, \quad z_5 = e^{i\cdot 5\pi/3}}$$

11. **(4 points)** Find the solution $y = y(x)$ to the following initial value problem:

$$y'' + 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 1$$

Solution:

The characteristic equation is:

$$r^2 + 2r + 5 = 0$$

Using the quadratic formula:

$$r = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = -1 \pm 2i$$

The roots are complex: $r = -1 \pm 2i$, so the general solution is:

$$y(x) = e^{-x} \left(c_1 \cos 2x + c_2 \sin 2x\right)$$

Apply initial condition $y(0) = 1$:

$$y(0) = c_1 = 1$$

The derivative is:

$$y'(x) = -e^{-x} \left(c_1 \cos 2x + c_2 \sin 2x\right) + e^{-x} \left(-2c_1 \sin 2x + 2c_2 \cos 2x\right)$$

Apply initial condition $y'(0) = 1$:

$$y'(0) = -c_1 + 2c_2 = 1 \implies -1 + 2c_2 = 1 \implies c_2 = 1$$

Answer: $\boxed{y(x) = e^{-x} \left(\cos 2x + \sin 2x\right)}$